

Problem 1 - Fourier Series

Compute the coefficients a_n, b_n in the Fourier series representations of each of the following functions. Then, determine the x at which the series representations fail, if any.

$$a_n = \frac{1}{L} \int_{-L}^L f(x) \sin\left(\frac{n\pi}{L}x\right) dx$$
$$b_n = \frac{1}{L} \int_{-L}^L f(x) \cos\left(\frac{n\pi}{L}x\right) dx$$

(a) $f(x) = 1 + \cos(x) - \sin(2x) \quad L = \pi$

$$\tilde{f}(x) = \frac{b_0}{2} + \sum_{n=1}^{\infty} a_n \sin(nx) + b_n \cos(nx)$$
$$1 + \cos(x) - \sin(2x) = \frac{b_0}{2} + \sum_{n=1}^{\infty} a_n \sin(nx) + b_n \cos(nx)$$

By inspection:

$$\frac{b_0}{2} = 1 \implies b_0 = 2$$
$$b_1 = 1, \quad b_n = 0 \text{ for } n \neq 1$$
$$a_2 = -1, \quad a_n = 0 \text{ for } n \neq 2$$

Since $f(x) = 1 + \cos(x) - \sin(2x)$ is smooth and continuous, its Fourier series representation does not fail for any x .

(b) $f(x) = x, \quad L = \pi$

Solving for a_n and b_n using formulas from class.

$$a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} x \sin(nx) dx$$

Integration by parts (table method):

x	$\sin(nx)$
-1	$-\frac{1}{n} \cos(nx)$
0	$-\frac{1}{n^2} \sin(nx)$

$$\begin{aligned}
a_n \pi &= -\frac{x}{n} \cos(nx) + \frac{1}{n^2} \sin(nx) \Big|_{-\pi}^{\pi} \\
a_n \pi &= -\frac{\pi}{n} \cos(n\pi) + \frac{1}{n^2} \sin(n\pi) - \left(-\frac{-\pi}{n} \cos(-n\pi) + \frac{1}{n^2} \sin(-n\pi) \right) \\
a_n \pi &= -\frac{\pi}{n} \cos(n\pi) - \left(-\frac{-\pi}{n} \cos(-n\pi) \right) \\
a_n &= -\frac{1}{n} \cos(n\pi) - \left(-\frac{1}{n} \cos(-n\pi) \right) \\
a_n &= -\frac{1}{n} \cos(n\pi) - \left(\frac{1}{n} \cos(-n\pi) \right) \quad \cos(n\pi) = \cos(-n\pi) = (-1)^n \\
a_n &= -\frac{1}{n} (-1)^n - \left(\frac{1}{n} (-1)^n \right) \\
a_n &= -\frac{2}{n} (-1)^n
\end{aligned}$$

Since $f(x) = x$ is an odd function, and $\cos(nx)$ is an even function, $f(x) \cos(nx)$ is an odd function. The integral $b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} x \cos(\frac{n\pi}{\pi} x) dx$ would be 0 since it is an odd function integrated over symmetric bounds. Therefore $b_n = 0 \forall n \in \mathbb{Z}^+$

The series fails at $x = \pi$ and $x = -\pi$ because of the jump discontinuities that exist there.

(c) $f(x) = e^x$, $L = \pi$

Using the formula for a_n :

$$a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} e^x \sin(n\pi) dx$$

Using integration by parts with $u = \sin(nx)$, $du = n \cos(nx)$, $dv = e^x$,
 $v = e^x$

$$\int_{-\pi}^{\pi} e^x \sin(n\pi) dx = \sin(nx)e^x - n \int e^x \cos(nx)$$

Using integration by parts again with $u = \cos(nx)$, $du = -n \sin(nx)$, $dv = e^x$, $v = e^x$

$$\begin{aligned} \int_{-\pi}^{\pi} e^x \sin(n\pi) dx &= \sin(nx)e^x - n \left(\cos(nx)e^x + n \int e^x \sin(nx) dx \right) \\ \int_{-\pi}^{\pi} e^x \sin(n\pi) dx &= \sin(nx)e^x - n \cos(nx)e^x - n^2 \int e^x \sin(nx) dx \\ \int_{-\pi}^{\pi} e^x \sin(n\pi) dx + n^2 \int_{-\pi}^{\pi} e^x \sin(nx) dx &= \sin(nx)e^x - n \cos(nx)e^x \Big|_{-\pi}^{\pi} \\ \int_{-\pi}^{\pi} e^x \sin(n\pi) dx (1 + n^2) &= \sin(nx)e^x - n \cos(nx)e^x \Big|_{-\pi}^{\pi} \\ \int_{-\pi}^{\pi} e^x \sin(n\pi) dx &= \frac{\sin(nx)e^x - n \cos(nx)e^x}{1 + n^2} \Big|_{-\pi}^{\pi} \end{aligned}$$

We know that $\cos(n\pi) = \cos(-n\pi) = (-1)^n$

$$\begin{aligned} \int_{-\pi}^{\pi} e^x \sin(n\pi) dx &= \frac{-n(-1)^n e^{\pi}}{1 + n^2} + \frac{n(-1)^n e^{-\pi}}{1 + n^2} \\ \int_{-\pi}^{\pi} e^x \sin(n\pi) dx &= \frac{-n(-1)^n}{1 + n^2} (e^{\pi} - e^{-\pi}) \\ a_n &= \frac{-n(-1)^n}{1 + n^2(\pi)} (e^{\pi} - e^{-\pi}) \end{aligned}$$

For b_n , we can also use the formula:

$$b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} e^x \cos(nx) dx$$

Using integration by parts, with $u = \cos(nx)$, $du = -n \sin(nx)$, $v = e^x$,
 $dv = e^x$.

$$\int_{-\pi}^{\pi} e^x \cos(nx) dx = e^x \cos(nx) + n \int e^x \sin(nx)$$

Using integration by parts again with $u = \sin(nx)$, $du = -n \cos(nx)$, $v = e^x$, $dv = e^x$.

$$\begin{aligned}
\int_{-\pi}^{\pi} e^x \cos(nx) dx &= e^x \cos(nx) + n \sin(nx) - n^2 \int e^x \cos(nx) \\
\int_{-\pi}^{\pi} e^x \cos(nx) dx + n^2 \int_{-\pi}^{\pi} e^x \cos(nx) dx &= e^x \cos(nx) + n \sin(nx) \Big|_{-\pi}^{\pi} \\
\int_{-\pi}^{\pi} e^x \cos(nx) dx (1 + n^2) &= e^x \cos(nx) + n \sin(nx) \Big|_{-\pi}^{\pi} \\
\int_{-\pi}^{\pi} e^x \cos(nx) dx &= \frac{e^x \cos(nx) + n \sin(nx)}{1 + n^2} \Big|_{-\pi}^{\pi} \\
\int_{-\pi}^{\pi} e^x \cos(nx) dx &= \frac{e^{\pi} \cos(n\pi)}{1 + n^2} - \frac{e^{-\pi} \cos(-n\pi)}{1 + n^2} \\
\int_{-\pi}^{\pi} e^x \cos(nx) dx &= \frac{e^{\pi} (-1)^n}{1 + n^2} - \frac{e^{-\pi} (-1)^n}{1 + n^2} \\
\int_{-\pi}^{\pi} e^x \cos(nx) dx &= \frac{(-1)^n}{1 + n^2} (e^{\pi} - e^{-\pi}) \\
b_n &= \frac{1}{\pi} \frac{(-1)^n}{1 + n^2} (e^{\pi} - e^{-\pi})
\end{aligned}$$

The series fails at $x = \pi$ and $x = -\pi$ because of the jump discontinuities that exist there.

$$(\mathbf{d}) \quad f(x) = \begin{cases} 0, & -\pi \leq x \leq 0, \\ 1 - \frac{x}{\pi}, & 0 < x \leq \pi \end{cases}, \quad L = \pi$$

Since the function is 0 on the interval $-\pi \leq x \leq 0$, our bounds for our integrals only have to go from 0 to π .

For b_0 :

$$\begin{aligned}
b_n &= \frac{1}{\pi} \int_0^{\pi} \left(1 - \frac{x}{\pi}\right) dx \\
&= \frac{1}{\pi} \left[x - \frac{x^2}{2\pi} \right]_0^{\pi} \\
&= \frac{1}{\pi} \left[\pi - \frac{\pi^2}{2\pi} \right] \\
b_0 &= 1 - \frac{1}{2} = \frac{1}{2}
\end{aligned}$$

For a_n :

$$a_n = \int_0^\pi \left(1 - \frac{x}{\pi}\right) \sin(nx) \, dx$$

Integration by parts using the table method:

$1 - \frac{x}{\pi}$	$\sin(nx)$
$\frac{1}{\pi}$	$-\frac{1}{n} \cos(nx)$
0	$-\frac{1}{n^2} \sin(nx)$

$$\begin{aligned} a_n &= \frac{1}{\pi} \left[-\left(1 - \frac{x}{\pi}\right) \frac{1}{n} \cos(nx) - \frac{1}{\pi n^2} \sin(nx) \right]_0^\pi \\ &= 0 - \left(\frac{-1}{n\pi}\right) a_n = \frac{1}{n\pi} \end{aligned}$$

For b_n :

$$b_n = \frac{1}{\pi} \int_0^\pi \left(1 - \frac{x}{\pi}\right) \cos(nx) \, dx$$

We can use integration by parts with the table method again:

$1 - \frac{x}{\pi}$	$\cos(nx)$
$\frac{1}{\pi}$	$\frac{1}{n} \sin(nx)$
0	$-\frac{1}{n^2} \cos(nx)$

$$\begin{aligned} b_n &= \frac{1}{\pi} \left[\left(1 - \frac{x}{\pi}\right) \frac{1}{n} \sin(nx) - \frac{1}{\pi n^2} \cos(nx) \right]_0^\pi \\ &= \frac{1}{\pi} \left[-\frac{1^n}{n^2 \pi} + \frac{1}{n^2 \pi} \right] \\ b_n &= \frac{1 - (-1)^n}{\pi^2 n^2} \end{aligned}$$

The series fails at $x = 0$ due to the jump discontinuity that exists there.

Problem 2 - Another Separation of Variables IBVP

Solve the finite-interval, **free-endpoint** wave equation IBVP given by

(a) Solve

$$\begin{cases} u_{tt} = c^2 u_{xx}, & x \in [0, 3], t \geq 0, \\ u_x(0, t) = 0, & t \geq 0, \\ u_x(3, t) = 0, & t \geq 0, \\ u(x, 0) = \cos^4\left(\frac{2}{3}\pi x\right), & x \in [0, 3], \\ u_t(x, 0) = 0, & x \in [0, 3]. \end{cases}$$

$$u_x(0, t) = u_x(3t) = 0$$

$$u(x, t) = \frac{b_0}{2} + \sum_{n=1}^{\infty} b_n \cos\left(\frac{n\pi}{3}x\right) \cos\left(\frac{cn\pi}{3}t\right)$$

$$u(x, 0) = \cos^4\left(\frac{2}{3}\pi x\right)$$

$$\text{Let } \theta = \frac{2}{3}\pi x$$

$$\cos^4(\theta) = (\cos^2(\theta))^2 = \left(\frac{1}{2}(1 + \cos(2\theta))\right)^2 \frac{1}{4}(1 + 2\cos(2\theta) + \cos^2(2\theta))$$

$$\frac{1}{4}(1 + 2\cos(2\theta) + \frac{1}{2} + \frac{1}{2}\cos(4\theta))$$

$$\frac{3}{8} + \frac{1}{2}\cos(2\theta) + \frac{1}{8}\cos(4\theta)$$

$$u(x, 0) = \frac{3}{8} + \frac{1}{2}\cos\left(\frac{4}{3}\pi x\right) + \frac{1}{8}\cos\left(\frac{8}{3}\pi x\right)$$

From inspection, we can see that:

$$n = 4, \quad a_4 = \frac{1}{2}$$

$$n = 8, \quad a_8 = \frac{1}{8}$$

$$n = 0, \quad a_0 = \frac{3}{4}$$

So the final equation is:

$$u(x, t) = \frac{3}{8} + \frac{1}{2} \cos\left(\frac{4\pi}{3}x\right) \cos\left(\frac{4c\pi}{3}t\right) + \frac{1}{8} \cos\left(\frac{8\pi}{3}x\right) \cos\left(\frac{8c\pi}{3}t\right)$$

(b) Plot

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Problem 3 - Laplace's Equation in Polar Coordinates

(a) Find ODEs

$$u(r, \theta) = R(r)\Theta(\theta)$$

$$u_r = R'(r)\Theta(\theta)$$

$$u_{rr} = R''(r)\Theta(\theta)$$

$$u_{\theta\theta} = R(r)\Theta''(\theta)$$

Substitute these into Laplace's equation:

$$R''\Theta + \frac{1}{r}R'\Theta + \frac{1}{r^2}R\Theta'' = 0$$

Separate the variables:

$$\begin{aligned} r^2 \frac{R''}{R} + r \frac{R'}{R} + \frac{\Theta''}{\Theta} &= 0 \\ r^2 \frac{R''}{R} + r \frac{R'}{R} &= -\frac{\Theta''}{\Theta} \end{aligned}$$

Since the left side depends only on r and the right side depends only on θ , both sides must be a constant λ .

$$\Theta'' + \lambda\Theta = 0$$

$$r^2 R'' + rR' - \lambda R = 0$$

(b) Solve the ODEs

Constraints:

$$\Theta(\theta + 2\pi) = \Theta(\theta)$$

$$\lim_{r \rightarrow 0} |R(r)| < \infty$$

Angular Equation: $\Theta'' + \lambda\Theta = 0$

From 2π -periodicity, $\lambda = n^2$ where $n = 0, 1, 2, \dots$

$$n = 0 : \quad \Theta_0(\theta) = a_0$$

$$n > 0 : \quad \Theta_n(\theta) = a_n \cos(n\theta) + b_n \sin(n\theta)$$

Radial Equation: $r^2 R'' + rR' - n^2 R = 0$

Let $R(r) = r^p$:

$$\begin{aligned} p(p-1) + p - n^2 &= 0 \\ p^2 - n^2 &= 0 \implies p = \pm n \end{aligned}$$

For $n = 0$, there is a repeated root $p = 0$:

$$\begin{aligned} R_0(r) &= C_0 + D_0 \ln(r) \\ \lim_{r \rightarrow 0} |R_0(r)| < \infty &\implies D_0 = 0 \\ R_0(r) &= C_0 \end{aligned}$$

For $n > 0$:

$$\begin{aligned} R_n(r) &= C_n r^n + D_n r^{-n} \\ \lim_{r \rightarrow 0} |R_n(r)| < \infty &\implies D_n = 0 \\ R_n(r) &= C_n r^n \end{aligned}$$

General solution by superposition:

$$u(r, \theta) = \frac{A_0}{2} + \sum_{n=1}^{\infty} r^n (A_n \cos(n\theta) + B_n \sin(n\theta))$$

(c) Boundary Value Problem

We are given the boundary condition at the edge of the disk ($r = R$):

$$\begin{aligned} u(R, \theta) &= \cos(\theta) \sin(\theta) \\ u(R, \theta) &= \frac{1}{2} \sin(2\theta) \quad (\text{using trig identities}) \end{aligned}$$

Evaluating our general solution at $r = R$ and setting it equal to this boundary condition:

$$\frac{A_0}{2} + \sum_{n=1}^{\infty} R^n (A_n \cos(n\theta) + B_n \sin(n\theta)) = \frac{1}{2} \sin(2\theta)$$

By inspection, we can match the orthogonal coefficients on both sides:

$$\begin{aligned} A_n &= 0 \quad \forall \quad n \\ B_n &= 0 \quad \text{for } n \neq 2 \\ R^2 B_2 &= \frac{1}{2} \implies B_2 = \frac{1}{2R^2} \end{aligned}$$

Plugging B_2 back into the general solution, the final exact solution is:

$$u(r, \theta) = \frac{1}{2R^2} r^2 \sin(2\theta)$$

Problem 4 - Operators and Inner Products

(a) Show that $\langle u, v \rangle_{L^2}$ defines an inner product.

To show that $\langle u, v \rangle_{L^2} = \int_{-1}^1 u(x)v(x) dx$ defines an inner product, we must verify three properties: symmetry, linearity, and positive definiteness.

1. **Symmetry:**

$$\begin{aligned} \langle u, v \rangle_{L^2} &= \int_{-1}^1 u(x)v(x) dx \\ &= \int_{-1}^1 v(x)u(x) dx \\ &= \langle v, u \rangle_{L^2} \end{aligned}$$

2. **Linearity:** For any scalars $a, b \in \mathbb{R}$ and functions $u, v, w \in C_0^2([-1, 1], \mathbb{R})$,

$$\begin{aligned} \langle au + bv, w \rangle_{L^2} &= \int_{-1}^1 (au(x) + bv(x))w(x) dx \\ &= \int_{-1}^1 (au(x)w(x) + bv(x)w(x)) dx \\ &= a \int_{-1}^1 u(x)w(x) dx + b \int_{-1}^1 v(x)w(x) dx \\ &= a \langle u, w \rangle_{L^2} + b \langle v, w \rangle_{L^2} \end{aligned}$$

3. **Positive Definiteness:**

$$\langle u, u \rangle_{L^2} = \int_{-1}^1 u(x)^2 dx$$

Since $u(x)^2 \geq 0$ for all real x , the integral over the domain is non-negative ($\langle u, u \rangle_{L^2} \geq 0$). Furthermore, since $u(x)$ is continuous, if $\int_{-1}^1 u(x)^2 dx = 0$, then $u(x) = 0$ everywhere on $[-1, 1]$. Thus, for any $u \neq 0$, $\langle u, u \rangle_{L^2} > 0$.

Thus, it satisfies all conditions and is an inner product.

(b) Show that $\mathcal{L} = \partial_x^2$ is self-adjoint.

By definition, $\mathcal{L}$ is self-adjoint if $\langle \mathcal{L}u, v \rangle = \langle u, \mathcal{L}v \rangle$. Evaluating the left side:

$$\langle \mathcal{L}u, v \rangle_{L^2} = \int_{-1}^1 u''(x)v(x) dx$$

We apply integration by parts, letting $w = v(x)$ and $dz = u''(x) dx$:

$$\int_{-1}^1 u''(x)v(x) dx = [u'(x)v(x)]_{-1}^1 - \int_{-1}^1 u'(x)v'(x) dx$$

Because $v \in C_0^2([-1, 1], \mathbb{R})$, we know the boundary conditions give $v(1) = v(-1) = 0$. Thus, the boundary term evaluates to 0:

$$\langle \mathcal{L}u, v \rangle_{L^2} = - \int_{-1}^1 u'(x)v'(x) dx$$

We apply integration by parts once more, this time letting $w = u'(x)$ and $dz = v'(x) dx$:

$$- \int_{-1}^1 u'(x)v'(x) dx = - \left([u(x)v'(x)]_{-1}^1 - \int_{-1}^1 u(x)v''(x) dx \right)$$

Because $u \in C_0^2([-1, 1], \mathbb{R})$, $u(1) = u(-1) = 0$, so the boundary term vanishes again:

$$\langle \mathcal{L}u, v \rangle_{L^2} = \int_{-1}^1 u(x)v''(x) dx = \langle u, \mathcal{L}v \rangle_{L^2}$$

Thus, $\mathcal{L} = \partial_x^2$ is self-adjoint.

(c) Show that eigenvalues are purely real and eigenvectors are orthogonal.

First, we show eigenvalues are purely real. Let $\mathcal{L}v = \lambda v$ for an eigenfunction $v \neq 0$. Taking the inner product of $\mathcal{L}v$ with v over the complex field

extension:

$$\langle \mathcal{L}v, v \rangle = \langle \lambda v, v \rangle = \lambda \langle v, v \rangle$$

By self-adjointness, $\langle \mathcal{L}v, v \rangle = \langle v, \mathcal{L}v \rangle$. Evaluating the right side:

$$\langle v, \mathcal{L}v \rangle = \langle v, \lambda v \rangle = \bar{\lambda} \langle v, v \rangle$$

Equating the two yields $\lambda \langle v, v \rangle = \bar{\lambda} \langle v, v \rangle$. Since $v \neq 0$, we know $\langle v, v \rangle > 0$, allowing us to divide by $\langle v, v \rangle$ to get $\lambda = \bar{\lambda}$. Therefore, λ is purely real.

Next, we show orthogonality. Let $\mathcal{L}u = \lambda_1 u$ and $\mathcal{L}v = \lambda_2 v$ where $\lambda_1 \neq \lambda_2$.

$$\begin{aligned} \lambda_1 \langle u, v \rangle &= \langle \lambda_1 u, v \rangle \\ &= \langle \mathcal{L}u, v \rangle \\ &= \langle u, \mathcal{L}v \rangle \quad (\text{by self-adjointness}) \\ &= \langle u, \lambda_2 v \rangle \\ &= \lambda_2 \langle u, v \rangle \end{aligned}$$

Subtracting the right side gives:

$$(\lambda_1 - \lambda_2) \langle u, v \rangle = 0$$

Since $\lambda_1 \neq \lambda_2$, we must have $\langle u, v \rangle = 0$, meaning the eigenfunctions associated with distinct eigenvalues are orthogonal.

(d) Show that $\mathcal{L} = \partial_x^2$ is negative definite.

We evaluate $\langle \mathcal{L}u, u \rangle_{L^2}$ for $u \neq 0$:

$$\langle \mathcal{L}u, u \rangle_{L^2} = \int_{-1}^1 u''(x)u(x) dx$$

Using integration by parts exactly as in part (b):

$$\int_{-1}^1 u''(x)u(x) dx = [u'(x)u(x)]_{-1}^1 - \int_{-1}^1 (u'(x))^2 dx$$

Since $u(1) = u(-1) = 0$, the boundary term drops out:

$$\langle \mathcal{L}u, u \rangle_{L^2} = - \int_{-1}^1 (u'(x))^2 dx$$

Because $(u'(x))^2 \geq 0$, the integral is always positive or zero. If it were exactly zero, $u'(x) = 0$ everywhere, implying $u(x)$ is a constant. The boundary conditions would force $u(x) = 0$, but we assumed $u \neq 0$. Thus, the integral is strictly positive, making the entire expression strictly negative:

$$\langle \mathcal{L}u, u \rangle_{L^2} < 0$$

This satisfies the definition of negative definite.

(e) Show that a negative-definite operator cannot have any positive eigenvalues.

Let λ be an eigenvalue with corresponding eigenfunction $v \neq 0$, so $\mathcal{L}v = \lambda v$. Taking the inner product with v :

$$\begin{aligned}\langle \mathcal{L}v, v \rangle &= \langle \lambda v, v \rangle \\ \langle \mathcal{L}v, v \rangle &= \lambda \langle v, v \rangle\end{aligned}$$

Since the operator is negative definite, we established that $\langle \mathcal{L}v, v \rangle < 0$. By the properties of an inner product, for $v \neq 0$, $\langle v, v \rangle > 0$. We can rearrange the equation to solve for λ :

$$\lambda = \frac{\langle \mathcal{L}v, v \rangle}{\langle v, v \rangle}$$

A strictly negative quantity divided by a strictly positive quantity is negative. Therefore, $\lambda < 0$, meaning the operator cannot have positive (or zero) eigenvalues.

(f) Find all eigenvalues and associated eigenfunctions of $\mathcal{L} = \partial_x^2$ on $C_0^2([-1, 1], \cdot)$.

We need to solve the spatial eigenvalue problem:

$$v''(x) = \lambda v(x) \quad \text{with boundary conditions} \quad v(-1) = 0, \quad v(1) = 0$$

From (e), we know $\lambda < 0$. Let $\lambda = -k^2$ for some real $k > 0$. The ODE becomes:

$$v''(x) + k^2 v(x) = 0$$

The general solution to this homogeneous ODE is:

$$v(x) = A \cos(kx) + B \sin(kx)$$

Applying our boundary conditions:

$$\begin{aligned}v(-1) &= A \cos(-k) + B \sin(-k) = A \cos(k) - B \sin(k) = 0 \\v(1) &= A \cos(k) + B \sin(k) = 0\end{aligned}$$

Adding the two equations yields $2A \cos(k) = 0$. Subtracting them yields $2B \sin(k) = 0$. Since we cannot have both $A = 0$ and $B = 0$ (which would give the trivial solution $v(x) = 0$), we have two independent sets of solutions:

Case 1: Even Eigenfunctions ($B = 0$ and $\cos(k) = 0$)

$$k = \frac{(2n-1)\pi}{2} \quad \text{for } n = 1, 2, 3, \dots$$

This yields the eigenfunctions and corresponding eigenvalues:

$$v_n(x) = \cos\left(\frac{(2n-1)\pi}{2}x\right), \quad \lambda_n = -\frac{(2n-1)^2\pi^2}{4}$$

Case 2: Odd Eigenfunctions ($A = 0$ and $\sin(k) = 0$)

$$k = n\pi \quad \text{for } n = 1, 2, 3, \dots$$

This yields the eigenfunctions and corresponding eigenvalues:

$$w_n(x) = \sin(n\pi x), \quad \lambda_n = -n^2\pi^2$$

Combined, these form the complete set of eigenvalues and eigenfunctions on the domain $[-1, 1]$ of length $L = 2$. Using a single index $m = 1, 2, 3, \dots$, we can concisely state the unified eigenvalues as:

$$\lambda_m = -\left(\frac{m\pi}{2}\right)^2$$

with the associated unified eigenfunctions:

$$u_m(x) = \sin\left(\frac{m\pi(x+1)}{2}\right)$$